SCHOOL OF ENGINEERING AND TECHNOLOGY ASSIGNMENT COVER SHEET

COURSE: NET1014 – Networking Principles

LEVEL: BCNS, BIT, BCS, BSE, BDS - Year 1

ACADEMIC SESSION: January 2024 Semester

DEADLINE: 17th Mar 2024

GROUP NO: _2_

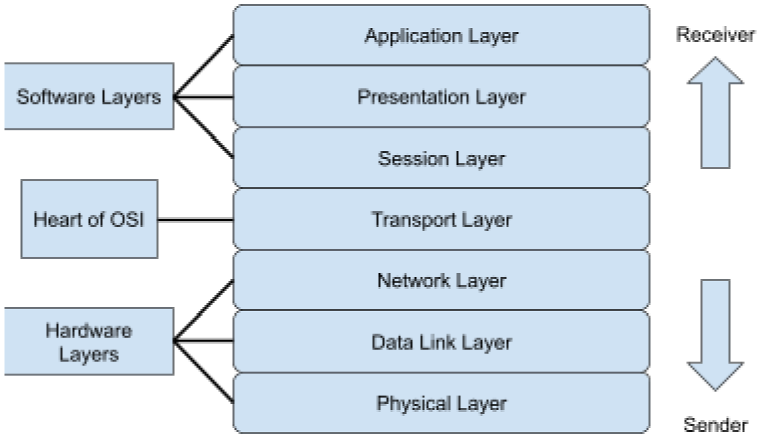
Part 1

1. Physical Layer
    - The Physical Layer transmits raw bit streams over the physical medium. It defines the hardware specifications, such as cables, connectors, voltage levels, and data rates.
    - Protocols: Ethernet, WI-FI, Fiber Distributed Data Interface
    - Tasks: Encoding, signaling, transmission of raw bits
    - Real-world Example: Ethernet is widely used in local area networks(LANs) for wired connections
2. Data Link Layer
    - This layer facilitates data transfer between 2 devices on the same network
    - It ensures that data is transmitted reliably across the physical medium
    - Protocols: Ethernet, Point-to-Point Protocol, High-level Data Link Control
    - Real-world Example: PPP is used for establishing direct connections between network nodes, commonly employed in dial-up connections
3. Network Layer
    - The Network Layer is responsible for facilitating data transfer between 2 different networks
    - It determines the optimal path for data packets to reach their destination
    - Protocols: Internet Protocol, Internet Control Message Protocol, Internet Group Management Protocol
    - Tasks: Logical addressing, routing, fragmentation, congestion control
    - Real-world Example: IP is the backbone of the internet, responsible for routing packets between networks
4. Transport Layer

- The Transport Layer ensures end-to-end communication between hosts and provides error checking and recovery
- It is responsible for flow control and error control
- Flow control determines an optimal speed of transmission, meanwhile error control ensures that the data receive is complete
- Protocols: Transmission Control Protocol, User Datagram Protocol
- Tasks: Segmentation, connection establishment, error recovery, flow control
- Real-world Example: TCP is used for reliable data transmission in applications like web browsing, email, and file transfer

5. Session Layer
    - The Session Layer responsible for opening and closing communication between the 2 devices, the time it opened and closed
    - It provides synchronization, checkpointing, and recovery services
    - Protocols: Remote Procedure Call, NetBIOS, Session Initiation Protocol
    - Tasks: Session establishment, synchronization, termination, data exchange
    - Real-world Example: SIP is used for initiating and terminating multimedia sessions, such as VoIP calls

6. Presentation Layer
    - The Presentation Layer handles data formatting, encryption, and compression to ensure compatibility between different systems
    - It makes the data presentable for applications to consume
    - Protocols: MIME(Multipurpose Internet Mail Extensions), SSL/LS,JPEG,GIF
    - Tasks: Data translation, encryption, compression, syntax conversion
    - Real-world Example: SSL/TLS provides secure communication over the internet, commonly used in online banking and e-commerce
7. Application Layer
    - The Application Layer is responsible for the protocols and data manipulation that the software relies on to present meaningful data to the user.
    - It enables communication between applications and supports various protocols for specific applications
    - Protocols: HTTP, FTP, SMTP, DNS, SSH, Telnet
    - Tasks: Application-specific protocols, data exchange, user interfaces
    - Real-world Example: HTTP is used for accessing web pages, while SMTP is used for sending emails



Advantages of the IOS Protocol Stack:
- Standardization: IOS is widely adopted, ensuring interoperability across different devices and networks
- Scalability: The modular design allows for easy integration of new technologies and protocols
- Robustness: The layered architecture provides fault isolation and facilitates troubleshooting

Potential Challenges:
- Security vulnerabilities require constant vigilance
- Complexity can lead to difficult troubleshooting
- Performance overhead due to protocol encapsulation

Part 2

Equipment List & Estimated Cost

Number of Equipment	Equipment List	Estimated Price per item (RM)	Total (RM)
	University Building		
1	Router 2911	2,620	2,620
	Lab (HUMAC)		
1	2960 switch	2,809	2,809
8	PC	2,099	16,792
8	Long Cable (copper straight-through)	537	4,296

1	Short Cable (copper straight-through)	51	51
	SET new building		
1	2960 switch	2,809	2,809
	Communications lab		
1	2960 switch	2,809	2,809
6	PC	2,099	12,594
6	Long Cable (copper straight-through)	537	3,222
1	Short Cable (copper cross-over)	52.33	52.33
	Advanced wireless lab		
1	2960 switch	2,809	2,809
4	PC	2,099	8,396
4	Long Cable (copper straight-through)	537	2,148
1	Short Cable (copper cross-over)	52.33	52.33
	Iot lab		
1	2960 switch	2,809	2,809
8	PC	2,099	16,792
8	Long Cable (copper straight-through)	537	4,296
1	Short Cable (copper cross-over)	52.33	52.33
	Between buildings		
1	Long Cable (copper straight-through)	537	537
			85,945.99

Device Justification

- 2911 Router:

In order to link various networks and enable communication between them, routers are necessary. The gateway that connects the university network to other networks, such the internet, is the Router 2911. In addition to provide network address translation (NAT), security features, and other capabilities necessary for internet connectivity, it controls traffic between the university's internal and external networks.

- 2960 Switches:

A number of equipment can be connected to a local area network (LAN) using switches. In order to provide connectivity between PCs and other devices, 2960 switches are utilised in every lab and building. They may manage the traffic produced within each lab or building as well as offer fast connectivity.

- Personal Computer (PC):

For a variety of tasks, including programming, research, simulations, and network resource access, personal computers are indispensable. Whether students, researchers, or faculty members, the number of PCs in each lab corresponds to the usual needs of those users.

- Straight-through copper cable:

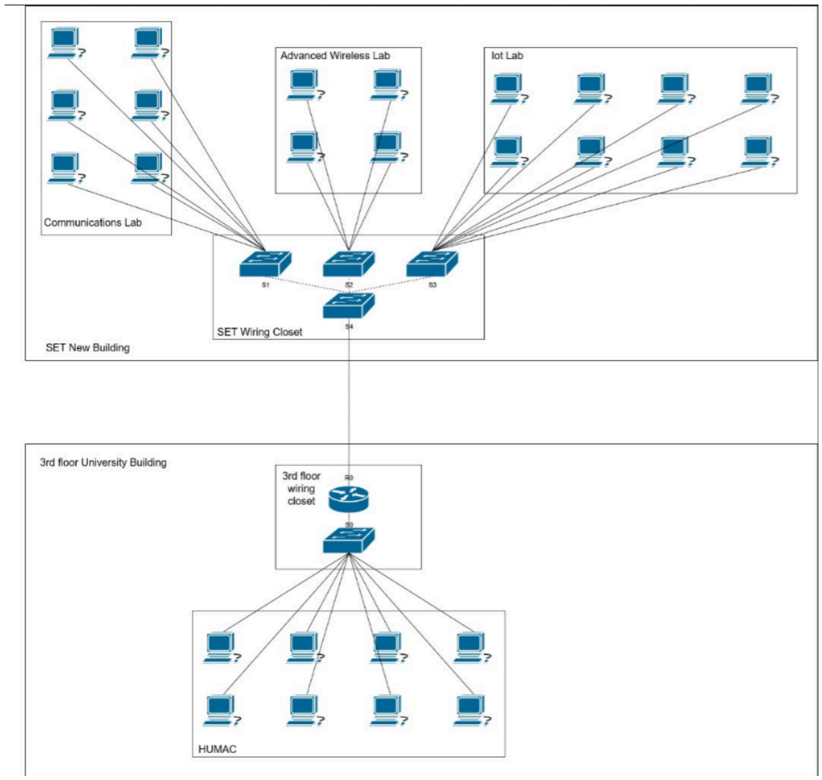
Straight-through connections are frequently used to join many device types together inside the same network segment, such as PCs and switches or switches and routers. In order to create wired connections between devices and networking equipment, these cables are widely used throughout the university building and labs.

- Cross-over (Copper Cable):

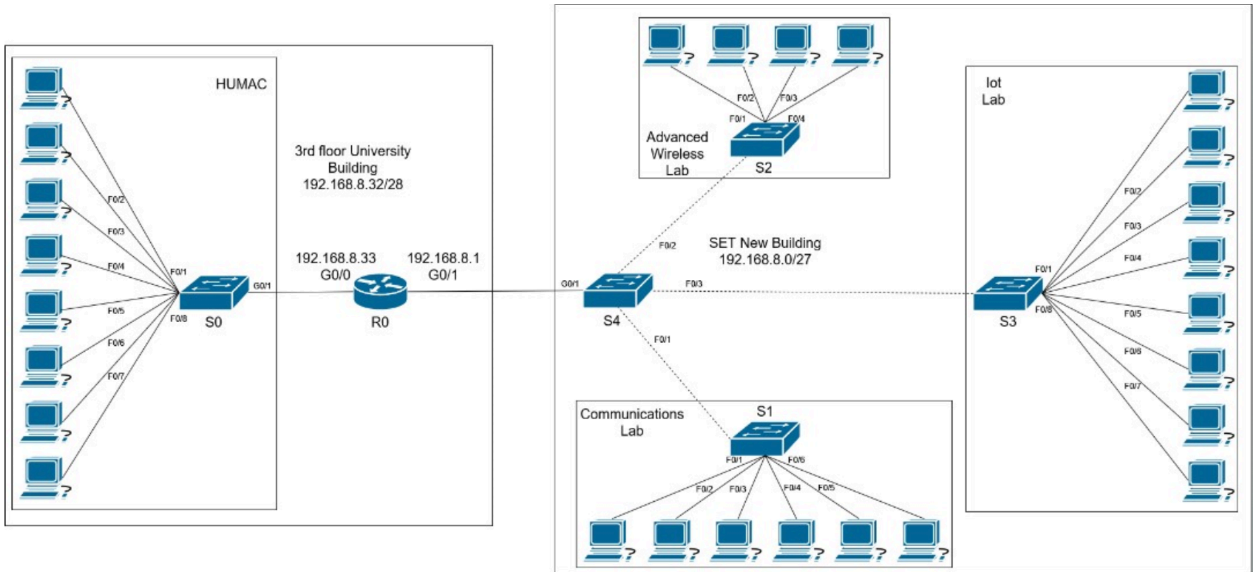
With crossover cables, similar devices can be connected directly to one another—for example, joining two switches together. They are rarely used but are essential in some situations, such when switches in the IoT, advanced wireless and communications lab need to be connected to a central switch.

To put it briefly, every piece of equipment and gadget has a distinct function in the creation and upkeep of the university's network infrastructure, as well as in enabling communication, supplying connectivity, and supporting a range of activities carried out in the labs and buildings.

Physical Topology



Logical Topology


HUMAC
3rd floor University
Building
192.168.8.32/28
192.168.8.33
G0/0
192.168.8.1
G0/1
S0
R0
G0/1
S4
SET New Building
192.168.8.0/27
Advanced
Wireless
Lab
S2
Iot
Lab
S3
Communications
Lab
S1
F0/1
F0/2
F0/3
F0/4
F0/5
F0/6
F0/7
F0/8